

Value co-creation for digital innovation: An interorganizational boundary-spanning perspective

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ABSTRACT

Despite the enthusiasm for engaging in interorganizational collaboration to enable digital product innovation, firms often face challenges in integrating knowledge across organizational boundaries. Our research examines how collaborating firms use Interorganizational Systems (IOS) tools and project coordinators to overcome knowledge boundaries in different types of ideation tasks. Drawing on boundary-spanning theory, we conceptualize IOS tools as boundary objects, project coordinators as boundary spanners, and the interdependence of ideation tasks as moderators of the impacts of boundary objects and spanners on knowledge integration. Our findings suggest that IOS tools help overcome syntax and semantic knowledge boundaries by transferring and sharing information when the interdependence of idea implementation tasks is high, and project coordinators help overcome pragmatic knowledge boundaries by building reciprocity of innovation participants when the interdependence of idea generation tasks is high. Our research contributes to value co-creation literature by developing a boundary-spanning mechanism to explain the roles of boundary objects and boundary spanners in overcoming different knowledge boundaries to enable digital product innovation at the inter-firm level.

1. Introduction

As digital technologies become highly distributed and change frequently, organizations have advanced from single-firm innovation to collaborative innovation across multiple firms [1]. Collaborative innovation focuses on the joint development of digital products by combining technical and business resources to generate knowledge across organizational boundaries [2,3]. Such innovation is exemplified in Samsung and Royal Philips teaming up for an intelligent Internet-of-Things healthcare platform [4] to Ericsson and Piaggio working on cloud-based connected vehicles [5]. Despite the prevalent examples in the industry, there is little scholarly investigation of how to co-create value for digital product innovation at the interorganizational level [6].

Interorganizational Systems (IOS) tools are used as “catalysts and enablers for collaboration across organizational boundaries and important drivers for value creation in inter-organizational networks” ([7], p. 288). Prior research focuses on the role of IOS tools in information sharing to improve operational efficiency between collaborative firms [8–10]. As well as sharing information, firms need to integrate domain-specific knowledge to create digital products [11,12].

Knowledge integration requires organizations to overcome syntactic boundaries by transferring information, semantic boundaries by developing the shared meaning of the information, and pragmatic boundaries by integrating information for new knowledge [13]. Research, to date, offers limited insights into how to use IOS tools to overcome these knowledge boundaries for digital innovation.

Interorganizational knowledge integration is challenging because collaborative firms have different organizational structures and cultures, causing barriers to sharing and synthesizing knowledge. In addition to IOS tools, firms need project coordinators to help increase the belongingness of team members and manage the conflict within boundary-spanning teams [14]. Moreover, the effects of IT and complementary resources on value co-creation are contingent on the interdependencies between collaborating firms i.e., high-level interdependencies positively moderate the relationship between IT assets and business value [15]. Focusing on the moderating effects of interdependencies, our research takes a step forward to examine if such effects vary across different types of tasks [16]. Taken together, we aim to develop a systematic understanding of how digital artifacts, humans, and interdependent tasks collectively contribute to the knowledge integration that leads to value co-creation for digital innovation.

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Our research focuses on ideation tasks including idea generation and idea implementation because they are two important stages in enabling digital innovation [17,18]. IOS tools and project coordinators enable digital innovation via interdependent tasks of generating and implementing ideas, and their roles may vary across these two types of tasks [16,19]. Therefore, we address the research question: how do IOS tools, project coordinators, and ideation task interdependence interact to integrate knowledge for digital product innovation?

Drawing on boundary-spanning theory, we conceptualize IOS tools as boundary objects and project coordinators as boundary spanners [13, 20]. Our research model examines 1) the enabling effects of boundary objects and boundary spanners on inter-firm knowledge integration that leads to high levels of digital innovation quality; and 2) the moderating effects of ideation task interdependence on the relationships between boundary resources (i.e., boundary objects; boundary spanners) and inter-firm knowledge integration.

We used the survey method to test our research model using data from 209 firms in China. The findings suggest that IOS tools and project coordinators are co-specialized assets for digital innovation. Specifically, coordinators are important for idea generation tasks because humans are effective in tacit knowledge integration, whereas IOS tools are critical for idea implementation tasks because these tools are effective in transferring codified, explicit knowledge. Our findings further suggest that IOS tools cannot directly lead to digital innovation quality and their effects are fully mediated by inter-firm knowledge integration, while boundary spanners can directly influence the quality of digital innovation.

This research makes three contributions to existing knowledge. First, it contributes to the IT-based value co-creation literature by developing a boundary-spanning mechanism from a knowledge development perspective [21,15]. The boundary-spanning mechanism provides a systematic understanding of how IOS tools, humans, and ideation tasks interact to integrate knowledge. IOS tools help organizations overcome syntactic and semantic knowledge boundaries when the interdependence of idea implementation tasks is high, while project coordinators help overcome pragmatic knowledge boundaries when the interdependence of idea generation tasks is high [22,13].

Second, our research contributes to the IOS literature by examining how IOS tools can be used for knowledge integration. Prior research largely focuses on the role of IOS tools in improving operational performance [10]. We examine IOS tools in knowledge-intensive activity i.e., enabling digital innovation, and suggest that the effects of IOS tools on knowledge integration are augmented when the information is presented in an explicit and codified format.

Third, our research contributes to the digital innovation literature by examining the enablement of digital product innovation at the inter-organizational level. Prior research largely focuses on the financial performance of digital innovation at a single firm [6]. Our research explains how to improve the quality of digital product innovation at the inter-firm level from a knowledge integration perspective.

2. Theoretical foundation

Because our research focuses on IOS-enabled value co-creation at the inter-firm level, we review the literature on IT-based value co-creation to highlight key issues to be addressed in this study. We then present boundary-spanning theory and explain how this research applies the theory to examine the inter-firm knowledge integration for digital innovation.

2.1. IT-based value co-creation

In the changing IT landscape featured by big data and ubiquitous technologies, developing digital products often requires collective intelligence across organizations. Firms, therefore, use IT to facilitate communication, accelerate innovation speed, and generate innovative

solutions [2,23]. IT-based value co-creation between firms emerges as an important research stream in the IS research field. This research stream, however, receives relatively less attention compared to the research that examines value creation at a single-firm level [7,15]. Specifically, IT-based value co-creation refers to using IT to generate business value at the inter-firm level [21,15].

We identify three issues in the literature on IT value co-creation and explain how our research addresses these issues. First, knowledge integration is an important aspect of value co-creation because collaborating firms need to share information and absorb external information to generate new knowledge at the interorganizational level [21,24]. This is especially important for digital innovation projects as it involves knowledge-intensive activities where participants generate innovative ideas and convert ideas into technical solutions [25,17]. Knowledge integration at the inter-firm level is challenging because innovation participants possess domain-specific knowledge and are bounded by their organizational routines and norms [13,26].

To effectively integrate domain-specific knowledge across organizational boundaries, firms need to not only overcome the difficulties of sharing information but also address the challenges of developing a shared vision and beliefs in developing new knowledge [27]. Prior research has not sufficiently addressed how IT and organizational resources work together to help overcome various knowledge boundaries (i.e., syntax, semantic, pragmatic) for innovation [13]. Furthermore, existing literature mainly examines the role of IOS in coordinating communication [9,28]. We lack the understanding of how IOS tools contribute to knowledge-driven value co-creation in interorganizational projects.

Second, existing literature lacks the investigation of the interdependence of tasks for value co-creation [15,16]. Interdependent tasks are channels where collaborating firms share information and resources to jointly integrate knowledge [16,29]. Prior research focuses on the interdependence of companies' business processes in supply chain management such as decision-making [30] and buyer-supplier relationships [31]. Unlike supply chain management, digital innovation relies on generating innovative ideas and converting these ideas into technical solutions [17,18]. As a result, the interdependence of idea generation and implementation tasks influence IOS tools and humans to co-create value. For example, idea generation tasks often deal with tacit knowledge. Because this knowledge resides in innovation actors' minds, the integration of such knowledge relies on human-to-human interactions [27]. Idea implementation tasks deal with explicit knowledge, which relies on IOS tools to share information [9]. However, there is limited research that sheds light on how IOS tools and project coordinators are used to engage in ideation tasks and how their roles may differ in different types of boundary-spanning tasks for innovation.

Finally, prior research mainly focuses on financial performance in general (e.g., sales; profits) or relationship-specific performance (e.g., shared trust; operational excellence) in supply chain management [7]. Few studies have examined IT-based value co-creation for digital products. Driven by the increasing importance of digital products, IS researchers have called for studies to examine how to enable digital product innovation [6]. Because emerging technologies such as artificial intelligence and the Internet of Things are highly distributed, knowledge to enable digital innovation is often derived from collaboration across multiple firms [11]. Therefore, it is useful to examine how to enable digital product innovation at the inter-firm level.

2.2. Boundary-spanning theory

Boundary-spanning theory suggests that organizations develop boundary objects, boundary spanners, and boundary-spanning tasks to achieve inter-functional collaboration within or across organizations [13,26,16]. Specifically, boundary objects refer to the artifacts that "are plastic enough to adapt to local needs and constraints of the several parties employing them, yet robust enough to maintain a common

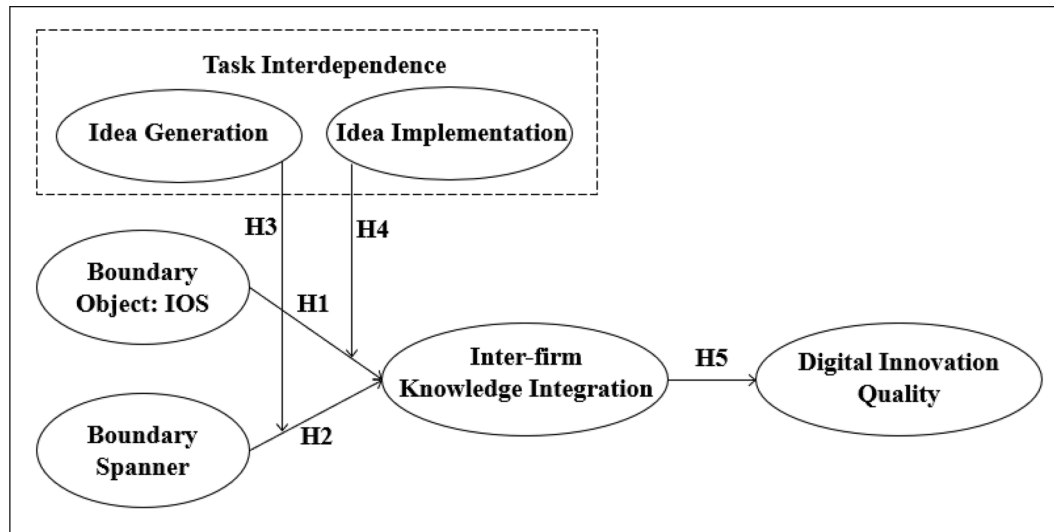


Fig. 1. Research Model of Value Co-creation for Digital Product Innovation.

identity across sites” ([32], p. 393). Boundary objects of IOS tools are exemplified by prototyping and computer-aided design systems, knowledge management systems, and collaboration systems [33,34]. Boundary spanners facilitate collaboration across different groups to overcome functional, geographical, or professional boundaries [26,11]. Boundary spanners are designated project coordinators who facilitate the collaboration between firms [13]. Boundary-spanning tasks involve interdependent activities that are collectively performed by collaborating organizations [16].

The IS research has examined the interactions of boundary spanners and boundary objects in shaping boundary-spanning activities within a firm or across firms. For example, boundary spanners facilitate the emergence of organizational boundary-spanning competence by coordinating communication, strengthening reciprocal relationships, and brokering knowledge [26,35]. Boundary objects such as collaboration tools help overcome knowledge boundaries by transferring, translating, and transforming information [13,22]. Ubiquitous technology provides opportunities for delivering new services by facilitating knowledge creation across organizational boundaries [36,11].

Prior research indicates that the roles of boundary spanners are contingent on the nature of the tasks [16,19]. Furthermore, researchers suggest that high-level task interdependence implies high knowledge barriers, which moderates the relationship between organizational capabilities (e.g., training, management support) and performance outcomes [29,37]. In line with these thoughts, boundary objects and spanners play different roles in shaping boundary-spanning activities, so their effects on knowledge integration are contingent on the interdependence of the tasks to be conducted [16].

In our research, we focus on the interdependence of ideation tasks because firms must generate ideas and convert these ideas into technical solutions to enable digital innovation. We examine idea generation and implementation tasks because they are two important stages of enabling digital product innovation [17,38]. Through interdependent idea generation and implementation tasks, collaborating firms jointly develop innovation. When the interdependence of idea generation tasks is high, boundary-spanning activities are characterized by tacit knowledge activities such as articulating ideas and developing a shared version of generating digital innovation. When the interdependence of idea implementation tasks is high, boundary-spanning activities are characterized by explicit knowledge activities such as sharing codified, explicit information across organizations. The interdependence of different types of ideation tasks, therefore, implies different knowledge barriers for boundary objects and spanners to engage in the process of integrating knowledge.

3. Research model and hypotheses development

Focusing on knowledge-driven value co-creation for digital innovation, we develop an integrative model to examine how IOS, project coordinators, and ideation task interdependence interact to integrate knowledge at the inter-firm level. Our model, as shown in Fig. 1, draws on the nomological framework of IT-based business value creation [15, 39] and boundary-spanning theory [22,20]. IT business value creation literature suggests that IS and organizational resources are antecedents to organizational performance [15,40,39]. Therefore, we propose that IOS tools and project coordinators are the antecedents to digital innovation quality.

Using boundary-spanning theory [13,22], we conceptualize IOS tools as boundary objects and project coordinators as boundary spanners. Boundary objects and boundary spanners enable knowledge integration at the inter-firm level, leading to high-level digital innovation quality. Boundary-spanning activities happen through interdependent idea generation and idea implementation tasks [17,16]. Highly coupled boundary-spanning activities are reflected in a high-level interdependence between tasks, and vice versa. The impacts of boundary objects and boundary spanners on knowledge integration are moderated by interdependent tasks where collaborating firms share information and build reciprocity [29,37]. Below, we present our five hypotheses in turn.

3.1. IOS and inter-firm knowledge integration

Boundary objects support the integration of knowledge across interorganizational boundaries syntactically, semantically, and pragmatically [13]. IOS tools are technological artifacts that are implemented to operationalize the relationships between collaborating organizations [7,33]. From the syntactic perspective, IOS tools increase collaborative organizations’ information processing capability by helping transfer knowledge across firms. For example, IOS tools provide a common platform where collaborators share the data and their understanding for solving problems and for discovering innovation opportunities [2]. Therefore, IOS tools support transferring domain-specific knowledge across syntactic boundaries [13].

From the semantic perspective, the use of IOS creates shared meanings, which allows collaborative organizations to translate knowledge. Because collaborators often come from complementary domains and have domain-specific knowledge, the shared meaning of the information from different domains is important for integrating knowledge. Collaborative technologies such as online communication tools or document-based collaboration systems facilitate the

interactions, which helps innovators make sense of semantic differences and smooth the process of knowledge integration [41].

From the pragmatic perspective, the use of IOS tools facilitates learning-by-doing processes where collaborators jointly transform their expertise into collective knowledge [13]. Individuals possess different skills that are specific to their local context, and knowledge gaps or boundaries may arise when collaborators apply domain-specific skills to developing digital innovation that involves cross-domain tasks [42]. To close these gaps, innovation actors need to experiment with their ideas and transform them into value-added output. Interorganizational prototyping systems such as computer-aided design systems (CAD) allow innovators to run and share simulations in trialing different conceptions for digital prototypes [33]. Hence, we propose the following hypothesis:

H1: *The use of IOS tools is positively associated with inter-firm knowledge integration.*

3.2. Boundary spanners and inter-firm knowledge integration

In collaborative innovation, coordination roles engage in meetings, face-to-face discussions, and seminars at different organizational levels [20]. Boundary-spanning coordinators help with knowledge integration in sharing information and coordinating collaboration activities. First, boundary spanners facilitate information flows across domains and bridge knowledge gaps from different fields. When innovators work on joint tasks, they face challenges in integrating local knowledge into the global level because these workers come from different firms and they may lack connections with other collaborators. Boundary spanners facilitate knowledge integration by establishing communication channels based on their formal and informal interactions with innovation participants. For example, working as knowledge brokers, boundary spanners identify knowledge gaps in newly joint groups and frame problems in the common language shared across functional teams [35].

Second, boundary spanners contribute to interorganizational knowledge integration by coordinating divergent views to reduce uncertainty and build reciprocal relationships among innovation participants. Frontline innovators often lack a global view because they mainly focus on the specific tasks assigned to them in a local context. Conflict or dispute may emerge when cross-boundary teams face competing tasks or requirements [26]. Project coordinators use formal and informal interactions to communicate and discuss uncertainties and articulate the vision of innovation development from a global perspective. These interactions build reciprocity between project members, facilitating the synthesis of complementary knowledge for innovation [11]. Hence, we propose the following hypothesis:

H2: *The use of boundary spanners is positively associated with inter-firm knowledge integration.*

3.3. Moderating effects of task interdependence

Task interdependence refers to the work processes that involve multiple actors who interact with collaborators to perform tasks shared across organizations [37]. In our research, we argue that the roles of boundary objects and boundary spanners are contingent on the interdependence of two types of ideation tasks. Specifically, we propose that the interdependence of idea generation tasks positively moderates the effects of boundary spanners on knowledge integration, and the interdependence of idea implementation tasks positively moderates the effects of boundary objects on knowledge integration.

Idea generation tasks focus on tacit knowledge because firms work on wicked problems at the early stage of innovation and the ideas are yet to take shape in an explicit, codified format [17]. For example, participants from a collaborative project interact to identify potential market needs, explore the requirements of potential customers, and discuss the shared vision of product development. The integration of the tacit knowledge possessed by individuals who are from different firms presents a high level of pragmatic knowledge boundaries as participants are

bounded by their organizational norms and interests [43]. Such kinds of integration rely on person-to-person interactions [44].

When the interdependence of idea generation tasks is high, innovation participants frequently interact to address the wicked problems by combining their knowledge to conceptualize key functionalities of digital innovation [17]. Frequent interactions strengthen the effects of boundary spanners on knowledge integration. When generating ideas, the project relies on boundary spanners to help overcome pragmatic boundaries by articulating a shared vision, negotiating potential conflict, and channeling communications [22].

Furthermore, project coordinators help shape the willingness of individuals to share their knowledge and increase the trust within the newly formed team to integrate knowledge [26]. For example, innovation participants from different domains may collect conflicting customer requirements in growing digital innovation. Under these situations, project coordinators reinforce the product development vision at the inter-organizational project level and configure different viewpoints in developing new knowledge to grow digital product innovation [23]. Hence, we propose the following hypothesis:

H3: *The interdependence of idea generation tasks positively moderates the effects of boundary spanners on inter-firm knowledge integration.*

During the idea implementation stage, the tacit knowledge about innovative ideas has been transformed into a codified format, which is ready to be converted to the functionalities of digital innovation. When the interdependence of idea implementation tasks is high, innovation participants frequently exchange information about realizing the functionalities of digital products [17]. The integration of explicit knowledge relies on digital artifacts that help overcome syntactic and semantic boundaries by developing a common lexicon and syntax [13]. The interdependence of idea implementation tasks, therefore, strengthens the effects of IOS tools on knowledge integration.

For example, when implementing ideas, collaborative teams use user case models to capture users' views that will be used to realize digital innovation based on market analysis and customer feedback [34]. IOS tools about case models provide a medium for firms to develop shared vocabularies and understanding to document customer requirements [42]. Furthermore, when the interdependence of idea implementation tasks is high, collaborative organizations tend to be overloaded by high-volume tasks. Under these circumstances, the effects of IOS tools are strengthened because the use of IOS tools allows innovators to exchange product data and associated information regardless of time and space. Hence, we propose the following hypothesis:

H4: *The interdependence of idea implementation tasks positively moderates the effects of IOS tools on inter-firm knowledge integration.*

3.4. Inter-firm knowledge integration and digital innovation quality

Inter-firm knowledge integration refers to the combination of individual know-how and expertise to achieve collaboration outcomes at the interorganizational level [45,46]. Interorganizational knowledge integration increases digital innovation quality in the following ways. First, a high level of knowledge integration implies a broad-scope synthesis of complementary knowledge for digital innovation [47]. Compared to cross-department integration for innovation within a firm, collaborative innovation often occurs through knowledge integration across industries. The greater the integration scope, the better the organization can offer appropriate and targeted product ideas [23]. For example, by integrating IoT technology from Samsung with a digital health platform from Royal Philips, these two firms developed an IoT-based platform where users can access, share, and analyze health data via connected devices [4].

Second, a high level of knowledge integration implies greater reconfiguration capabilities for generating innovation. Compared to traditional products, digital products offer fluid boundaries where changes happen frequently because technology is highly distributed and readily reprogrammable [48]. Superior knowledge integration between

Table 1
Operationalization of Constructs.

Variables	Item Description	References
<i>Interorganizational Systems (IOS)</i>	For the collaborative innovation project identified, please rate your extent of usage of the following interorganizational system tools. IOS1: Shared digital storage tools IOS2: Project collaboration tools IOS3: Prototyping tools IOS4: CAD modeling tools	[13,33], and Interviews with project managers
<i>Boundary Spanners (BS)</i>	For the collaborative innovation project identified, please rate your agreement on the following statements. BS1: One or more members of the project team had been formally designated to facilitate coordination with our collaborator. BS2: One or more members of the project team informally facilitated coordination with our collaborator. BS3: One or more members of the project team had been formally designated to facilitate knowledge sharing with our collaborator. BS4: One or more members of the project team informally facilitated knowledge sharing with our collaborator.	[20] Adapted from discussion in [26]
<i>Idea Generation Task Interdependence (IGTI)</i>	For the collaborative innovation project identified, please rate your agreement on the following statements <u>when generating ideas for innovation</u> . IGTI1: When generating ideas for innovation, our tasks can be performed fairly independently of our collaborator's tasks. (Reverse Code) IGTI2: When generating ideas for innovation, our tasks can be planned with little need to coordinate with our collaborator's tasks. (Reverse Code) IGTI3: When generating ideas for innovation, we rarely obtained information from our collaborators to complete our tasks. (Reverse Code) IGTI4: When generating ideas for innovation, our tasks were relatively unaffected by the performance of our collaborators. (Reverse Code) IGTI5: When generating ideas for innovation, our tasks required frequent coordination with the efforts of our collaborators. IGTI6: When generating ideas for innovation, our performance on tasks depends on receiving accurate information from our collaborators.	[29,37]
<i>Idea Implementation Task Interdependence (IITI)</i>	For the collaborative innovation project identified, please rate your agreement on the following statements <u>when implementing ideas for innovation</u> . IITI1: When implementing ideas for innovation, our tasks can be performed fairly independently of our collaborator's tasks. (Reverse Code) IITI2: When implementing ideas	[29,37]

Table 1 (continued)

Variables	Item Description	References
	for innovation, our tasks can be planned with little need to coordinate with our collaborator's tasks. (Reverse Code) IITI3: When implementing ideas for innovation, we rarely obtained information from our collaborators to complete our tasks. (Reverse Code) IITI4: When implementing ideas for innovation, our tasks were relatively unaffected by the performance of our collaborators. (Reverse Code) IITI5: When implementing ideas for innovation, our tasks required frequent coordination with the efforts of our collaborators. IITI6: When implementing ideas for innovation, our performance on tasks depends on receiving accurate information from our collaborators.	
<i>Inter-firm Knowledge Integration (IKI)</i>	For the collaborative innovation project identified, please rate your agreement on the following statements. IKI1: We combined our expertise with our collaborator's expertise to jointly solve project-related problems. IKI2: We combined our perspectives with our collaborator's perspectives to develop shared concepts for the digital product. IKI3: We gained new insights by sharing our ideas with our collaborators. IKI4: We shared our knowledge with collaborators to improve work efficiency. IKI5: We synthesized knowledge by combining expertise from collaborators. IKI6: We used the information from collaborators when making project-related decisions.	[45,46]
<i>Digital Innovation Quality (DIQ)</i>	For the collaborative innovation project identified, please rate your agreement on the following statements. DIQ1: Compared with technically similar, self-developed products, the technical capabilities of this digital product were better. DIQ2: Compared with competitors, the technical capabilities of this digital product were better. DIQ3: Compared with competitors, the features of this digital product were better. DIQ4: Our customers more often praised this digital product's quality. DIQ5: Our customers were satisfied with this digital product's quality DIQ6: Our customers were firmly convinced that the quality of this digital product is high.	[34,54,55]

collaborative organizations can help assimilate relevant technology and transform customer requirements into new products [2]. For example, by combining Pfizer's scientific knowledge and IBM's machine learning and data analytics technologies, the two firms develop a cloud-based health analytics platform, aiming to provide real-time, evidence-based insights for cancer treatment [49]. Hence, we propose the following hypothesis:

H5: *Inter-firm knowledge integration is positively associated with digital innovation quality.*

4. Research method

We test our hypotheses using a field survey. The unit of the analysis is an interorganizational, digital innovation project. In this section, we first present our measurement development, followed by the approaches that we use to validate our constructs. Finally, we present the procedures for collecting data.

4.1. Measurement development

We operationalize constructs in the survey instrument by drawing on validated measurement scales from the literature with a 7-point Likert response scale. We operationalized the dependent variable (digital innovation quality), independent variables (boundary spanners, inter-firm knowledge integration) and moderating variables (idea generation task interdependence, idea implementation task interdependence) as reflective constructs.

We operationalized the independent variable IOS tools as a formative construct [50]. For its measurements, we listed a number of IOS tools used in innovation projects based on existing literature [13,33]. To confirm and refine our list of tools, we conducted interviews with nine project managers in three firms to learn about the IOS tools that have been adopted in their projects (see Appendix A for interview details). Following the suggestions from our interviewees, we replaced 'databases' with 'shared digital storage' because the latter is commonly used in the industry. Using existing literature and the information collected from project managers, the items 'shared digital storage tools', 'prototyping tools', 'project collaboration tools', and 'CAD modeling tools' form the construct IOS tools.

To exclude alternative explanations, we include control variables firm age, size, and R&D investment because they indicate a firm's internal R&D characteristics and have impacts on innovation outcomes [51,52]. We also include industry as a control variable to account for the external environment where a firm operates [51]. Furthermore, we include prior relationships, project size, and project cost as project-related characteristics, which may influence collaborative innovation project outcomes [53]. Table 1 presents the operationalization of our constructs.

4.2. Conceptual validation

We consulted several colleagues in the IS research field to identify and revise the framing issues of our survey instrument. We developed measures based on the existing literature and validated our measures following Moore and Benbasat's [56] approach. Specifically, we first conducted a two-stage Q-sorting to verify the content validity, convergent validity, and discriminant validity of our measurements. The accuracy at the unlabeled stage is 90%, while the accuracy at the labeled stage is 92%. The Q-sorting results suggest good measurement quality. Based on the verification, we made minor modifications. We developed our instrument in English and then translated it into Chinese. To ensure conceptual equivalence, we used the back-translation approach [57]. To check face validity, we invited the top managers of three firms to comment on the survey. Based on their suggestions, we further phrased and framed the questions of the survey instrument.

Table 2
Demographics of Respondents ($N = 209$).

Control Variables		Frequency	Percentage
Industry	Industry 1: Chemical and pharmaceutical	38	18.2
	Industry 2: Industrial and commercial machinery equipment	43	20.6
	Industry 3: Electronic and other electrical equipment	49	23.4
	Industry 4: Fabricated metal and other material	46	22.0
	Others	33	15.8
	< 100	27	12.9
Number of Employees	100–300	87	41.6
	301–500	43	20.6
	501–1000	26	12.4
	1001–2000	12	5.7
	> 2000	14	6.7
Firm Age (years)	< 15	83	39.7
	15–20	83	39.7
	20–25	22	10.5
	> 25	21	10.0

4.3. Data collection procedures

We distributed our survey to 758 organizations that were randomly selected from a high-tech organization directory in China. We collected data from project managers who are in charge of inter-firm innovation projects by using the key informant approach [58]. We asked project managers to identify a collaborative innovation project and respond to our survey questions. Of the 232 returned surveys, we removed 23 surveys because of the incompleteness of data. We used 209 surveys for the data analysis. The sample size was sufficient to achieve a statistical power higher than 0.9 [59].

Table 2 presents the demographic characteristics of the respondents that we use as our control variables. Our target population consists of firms that have collaborated with other firms to co-develop digital products using IOS tools. We followed the procedures used by prior research to ensure our sample is representative of the target population. Specifically, we selected four industries that have a broad presence in China's economy and represent the most advanced high-tech development in China [60]. Furthermore, we selected firms that are of different ages and sizes to ensure the diversity of our sample [3]. We assess nonresponse bias by comparing the earliest 52 respondents (25%) and the last 52 respondents (25%) on their organizational characteristics and their responses to the principal constructs [61]. There are no statistical differences between the two cohorts. Nonresponse bias, therefore, is not a major concern in our study.

5. Analysis and results

We used variance-based partial least squares (PLS-SEM) to estimate the structural equation model [62,63]. Compared with covariance-based SEM, PLS-SEM is especially suitable for theory development and for a model including both reflective and formative constructs [64,65]. Our research focuses on developing a theoretical model of value co-creation for digital innovation and the model includes both reflective and formative constructs, so PLS-SEM is a suitable method to test our model.

In this section, we first demonstrate the reliability and validity of our measurement model. We then present the results for our hypotheses testing, followed by our mediation analysis results. Finally, we test our non-hypothesized relationships to investigate if 1) the interdependence of idea generation tasks moderates the effects of IOS on inter-firm knowledge integration; and 2) the interdependence of idea implementation tasks moderates the effects of boundary spanners on inter-firm integration.

Table 3

Loadings and Cross-loadings of the Reflective Constructs.

Items	BS	IGTI	IITI	IKI	DIQ
Boundary Spanners (BS)					
BS1	0.927	0.128	0.227	0.419	0.371
BS2	0.937	0.048	0.214	0.455	0.358
BS3	0.889	0.088	0.279	0.349	0.268
BS4	0.875	0.051	0.136	0.368	0.332
Idea Generation Task Interdependence (IGTI)					
IGTI1	0.080	0.955	0.131	−0.154	−0.106
IGTI2	0.091	0.961	0.129	−0.141	−0.133
IGTI3	0.073	0.968	0.116	−0.181	−0.127
IGTI4	0.095	0.957	0.137	−0.118	−0.105
IGTI5	0.069	0.968	0.084	−0.172	−0.139
IGTI6	0.099	0.957	0.127	−0.145	−0.092
Idea Implementation Task Interdependence (IITI)					
IITI1	0.182	0.109	0.948	0.100	−0.137
IITI2	0.242	0.084	0.922	0.102	−0.118
IITI3	0.272	0.092	0.904	0.117	−0.118
IITI4	0.235	0.136	0.949	0.132	−0.078
IITI5	0.167	0.138	0.954	0.102	−0.132
IITI6	0.180	0.134	0.864	0.058	−0.123
Inter-firm Knowledge Integration (IKI)					
IKI1	0.451	−0.127	0.265	0.836	0.413
IKI2	0.399	−0.080	0.321	0.762	0.300
IKI3	0.469	−0.086	0.301	0.817	0.386
IKI4	0.284	−0.167	−0.044	0.784	0.528
IKI5	0.256	−0.162	−0.170	0.775	0.612
IKI6	0.195	−0.144	−0.196	0.757	0.581
Digital Innovation Quality (DIQ)					
DIQ1	0.386	−0.080	−0.136	0.560	0.924
DIQ2	0.387	−0.067	−0.149	0.569	0.926
DIQ3	0.321	−0.189	−0.096	0.549	0.935
DIQ4	0.320	−0.105	−0.062	0.522	0.938
DIQ5	0.330	−0.107	−0.100	0.553	0.930
DIQ6	0.300	−0.145	−0.143	0.523	0.911

5.1. Measurement model: reliability and validity

To assess the reliability of reflective measures, we tested indicator reliability, construct reliability, convergent validity, and discriminant validity [66,65]. As shown in Table 3, all indicators achieve satisfactory reliability with outer loadings exceeding 0.70. The reflective measurement models present reliable internal consistency by achieving composite reliability of 0.908 or higher. The internal consistency is further evidenced by the average variance extracted (AVE) values, exceeding the threshold value of 0.5 [67,68].

To examine the discriminant validity of constructs, we used two measures: the indicators' cross-loadings [62] and the criteria developed by Fornell and Larcker [67]. Both cross-loadings (as shown in Table 3) and the comparison of the square roots of AVE and inter-construct correlations with other constructs (as shown in Table 4) demonstrate discriminant validity. We further assess the discriminant validity using the Heterotrait-Monotrait ratio (HTMT) criterion in Table 5. All values were below the conservative threshold value of 0.85 [63], reconfirming the established discriminant validity.

We followed the guidelines of Cenfetelli and Bassellier [69] and Petter et al. [50] to validate the formative construct of boundary objects: IOS tools. Table 6 shows the results for VIF (multicollinearity test),

Table 4

Internal Consistency, Convergent Validity, and Discriminant Validity for Reflective Constructs.

Latent Construct	Cronbach's α	Composite Reliability	AVE	Fornell-Larcker Criterion				
				BS	IGTI	IITI	IKI	DIQ
Boundary Spanners (BS)	0.928	0.949	0.823	0.907				
Idea Generation Task Interdependence (IGTI)	0.983	0.986	0.923	0.087	0.961			
Idea Implementation Task Interdependence (IITI)	0.966	0.972	0.854	0.235	0.124	0.924		
Inter-firm Knowledge Integration (IKI)	0.879	0.908	0.623	0.442	−0.161	0.115	0.789	
Digital Innovation Quality (DIQ)	0.968	0.974	0.860	0.369	−0.123	−0.124	0.590	0.927

Note: Square roots of AVE on the diagonal (in bold) and correlations among the latent constructs on the off-diagonal positions.

weights (the relative contribution), and loadings (the absolute importance). The assessment output suggests that multicollinearity is not an issue. Because we use single informants to answer our survey questions, we further assess the common method variance of our research in Appendix C. The assessment shows that our results are not subject to the issue of common method variance.

5.2. Hypotheses testing

We used the following criteria to evaluate the structural model: the squared multiple correlations R^2 , structural path coefficients, the predictive relevance of endogenous variables using Stone Geisser's Q^2 values, and effect size values f^2 as recommended by Hair et al. [63]. When performing the analysis, we selected the path-weighting scheme and a stop criterion set at 10^{-7} , with 5000 iterations of resampling and the no-sign-change using SmartPLS 3. We assessed all statistical tests at the two-tailed 5% significance level.

To reduce the threats of multicollinearity, we used mean centering before creating the interaction terms [70]. Furthermore, VIF values are fewer than the recommended threshold of 3.3 (the highest VIF among the explanatory variables was 3.188) [71]. The effect size values f^2 show small to moderate effects, supporting the predictive relevance of the endogenous latent variables. Table 7 presents the results of the hypothesized relationships.

Our results in Fig. 2 show that the conceptual model explains 47.9% of the variation in inter-firm knowledge integration and 43.3% of the variation in digital innovation quality. The results suggest that boundary objects are significantly and positively related to inter-firm knowledge integration ($\beta = 0.280, p < 0.001$), supporting H₁. Boundary spanners are significantly and positively related to inter-firm knowledge integration ($\beta = 0.163, p < 0.05$), supporting H₂. Finally, inter-firm knowledge integration is also statistically significant in explaining

Table 5

Heterotrait-Monotrait Ratio (HTMT) Assessment for Discriminant Validity.

Latent Construct	BS	IGTI	IITI	IKI	DIQ
Boundary Spanners (BS)					
Idea Generation Task Interdependence (IGTI)	0.093				
Idea Implementation Task Interdependence (IITI)	0.244	0.130			
Inter-firm Knowledge Integration (IKI)	0.476	0.171	0.297		
Digital Innovation Quality (DIQ)	0.384	0.126	0.131	0.645	

Table 6

Assessment of Formative Construct - IOS.

Items	Variance Inflation Factors	Weights	Loadings
IOS1*	3.202	0.346	0.917
IOS2	3.017	0.167	0.867
IOS3	2.836	0.329	0.903
IOS4	3.188	0.268	0.902

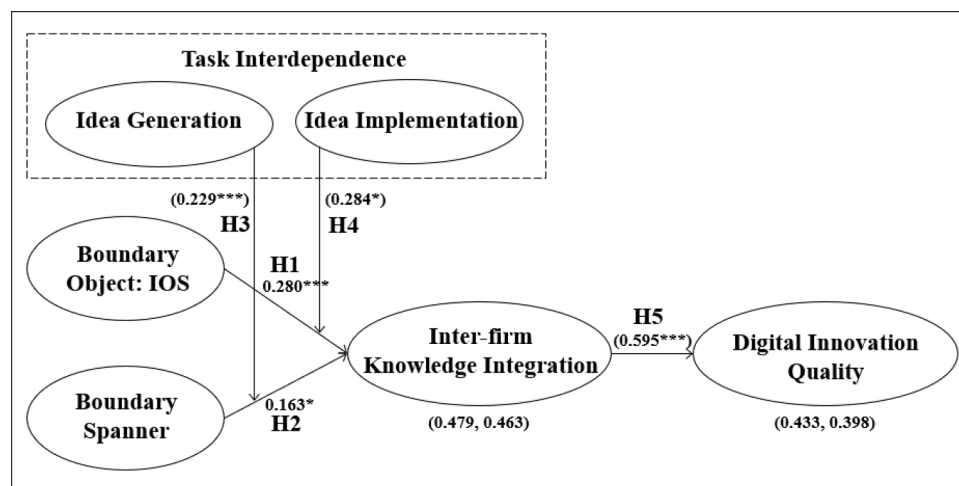
* See Table 1 for the description of items.

Table 7

Results of the Structural Model Path Coefficients.

Structural Path	Path Coefficient (β)	Effect Size (f^2)	Standard Deviation	t-value	Bias Corrected 95% Confidence Interval	Conclusion
IOS→IKI	0.280***	0.111	0.059	4.745	[0.150; 0.390]	H ₁ supported
BS→IKI	0.163*	0.032	0.067	2.456	[0.024; 0.283]	H ₂ supported
BS×IGTI→IKI	0.229***	0.091	0.058	3.949	[0.109; 0.324]	H ₃ supported
IOS×IITI→IKI	0.284**	0.117	0.122	2.336	[0.152; 0.411]	H ₄ supported
IKI→DIQ	0.595***	0.594	0.058	10.168	[0.454; 0.694]	H ₅ supported
Firm Age→DIQ	0.106	0.017	0.063	1.666	[-0.035; 0.214]	Not supported
Firm Size→DIQ	-0.056	0.003	0.079	0.704	[-0.226; 0.085]	Not supported
Industry1→DIQ	-0.059	0.003	0.070	0.836	[-0.195; 0.077]	Not supported
Industry2→DIQ	-0.051	0.002	0.075	0.677	[-0.196; 0.100]	Not supported
Industry3→DIQ	0.014	0.000	0.070	0.205	[-0.121; 0.151]	Not supported
Industry4→DIQ	0.111	0.011	0.067	1.659	[-0.019; 0.239]	Not supported
R&D→DIQ	-0.102	0.009	0.080	1.273	[-0.262; 0.056]	Not supported
Relationship→DIQ	0.077	0.008	0.063	1.223	[-0.039; 0.212]	Not supported
Project Size →DIQ	0.104	0.013	0.065	1.610	[-0.020; 0.238]	Not supported
Project Cost→DIQ	0.007	0.000	0.065	0.114	[-0.119; 0.126]	Not supported

* Significant at the 0.05 level; ** Significant at the 0.01 level; *** Significant at the 0.001 level.

**Fig. 2.** Model Estimation*Significant at the 0.05 level; **Significant at the 0.01 level; *** Significant at the 0.001 level. (R^2 adjusted, Q^2) given for endogenous constructs.**Table 8**

Results of Mediation Analysis (Mediator: Inter-firm Knowledge Integration).

Path	Direct Effect	95% Confidence Interval of the Direct Effect	t-value	Indirect Effect	95% Confidence Interval of the Indirect Effect	t-value	Conclusion
IOS→DIQ	0.171*	[0.079; 0.355]	2.389	0.058	[-0.132; 0.202]	0.709	Full mediation
BS→DIQ	0.228**	[0.001; 0.290]	3.253	0.141*	[0.024; 0.284]	2.124	Partial mediation
IOS×IITI→DIQ	0.198**	[0.073; 0.358]	2.645	0.077	[-0.061; 0.207]	1.119	Full mediation
BS×IGTI→DIQ	0.164*	[0.022; 0.289]	2.503	0.059	[-0.039; 0.172]	1.076	Full mediation

Note. * Significant at the 0.05 level; ** Significant at the 0.01 level; *** Significant at the 0.001 level.

digital innovation quality ($\beta = 0.595, p < 0.001$), supporting hypothesis H₅.

Our results suggest that the interaction effect of boundary spanners and idea generation task interdependence positively affects inter-firm knowledge integration ($\beta = 0.229, p < 0.001$), thereby supporting H₃. Furthermore, idea implementation task interdependence significantly moderates the effect of boundary objects on inter-firm knowledge integration ($\beta = 0.284, p < 0.05$), thereby supporting H₄. There is no significant influence of the control variables (i.e., firm age, size, industry, and other project-level control variables) on digital innovation quality.

5.3. Mediation analysis

To further check the mediation effect of inter-firm knowledge integration, we used independent variables to predict the dependent variable (i.e., digital innovation quality). Table 8 summarizes the mediation analysis. The results show the direct effects of boundary spanners ($\beta = 0.228, p < 0.01$), boundary objects ($\beta = 0.171, p < 0.05$), boundary spanners x idea generation task interdependence ($\beta = 0.164, p < 0.05$), and boundary objects x idea implementation task interdependence ($\beta = 0.198, p < 0.01$) are positively associated with digital innovation quality.

We also analyzed independent variables and a mediator predicting the dependent variable. The results show the effects of boundary objects ($\beta = 0.058, p > 0.05$), boundary spanners x idea generation task

interdependence ($\beta=0.059$, $p > 0.05$), and boundary objects $\times$ idea implementation task interdependence ($\beta = 0.077$, $p > 0.05$) with the mediation of inter-firm knowledge integration are no longer significantly associated with digital innovation quality. Taken together, the findings support the full mediation of inter-firm knowledge integration on boundary objects and the two interaction terms. In addition, the effect of boundary spanners remains significant, while weakened ($\beta = 0.141$, $p < 0.05$), which suggests the partial mediation of inter-firm knowledge integration on boundary spanners.

5.4. Post hoc analysis

We performed a post hoc analysis to examine the non-hypothesized moderating relationships between boundary objects and spanners and task interdependence. These moderating effects are (1) idea implementation task interdependence moderates the relationship between boundary spanners and inter-firm knowledge integration; and (2) idea generation task interdependence moderates the relationship between boundary objects and inter-firm knowledge integration. First, the interaction of boundary spanners and idea implementation task interdependence does not significantly influence inter-firm knowledge integration ($\beta = -0.020$, $p > 0.05$). Second, the interaction of boundary objects and idea generation task interdependence does not have a significant impact on inter-firm knowledge integration ($\beta = 0.009$, $p > 0.05$).

Because the interaction effects of boundary spanners and idea implementation tasks and the interaction effects of boundary objects and idea generation tasks are insignificant, our post hoc analysis results strengthen the finding that boundary objects and boundary spanners are co-specialized assets in integrating knowledge. In other words, the effects of IOS tools on knowledge integration are augmented only when the interdependence of idea implementation tasks is high, and the effects of project coordinators on knowledge integration are augmented only when the interdependence of idea generation tasks is high. Therefore, we suggest that boundary spanners are important for generating ideas that deal with tacit knowledge, whereas boundary objects are crucial for implementing ideas that deal with explicit knowledge.

6. Discussion and conclusion

In this section, we discuss the data analysis results, followed by the theoretical contributions of our research. We then present our research implications to practice, its limitations and opportunities for future research.

6.1. Discussion of findings

In this study, we develop a boundary-spanning mechanism to explain how IOS tools and boundary spanners lead to knowledge-driven value co-creation for digital product innovation. Our findings suggest that project coordinators can directly influence digital innovation quality, whereas IOS tools cannot. Prior research has suggested that humans are crucial for sensing innovation opportunities, formulating problems, and converting ideas into technical solutions [27,18]. Our research takes these findings further by demonstrating that human agents such as project coordinators have direct effects on the quality of digital products.

The effects of IOS tools on digital innovation quality are fully mediated by knowledge integration. These findings align with the IS business value literature in general, suggesting that the strategic value of IT assets is realized via intermediated processes [39]. Our research specifically demonstrates that interorganizational knowledge integration capabilities are critical for IOS tools to create value for digital product innovation. Given that digital innovation involves knowledge-intensive activities, IOS tools facilitate the combination of

domain-specific knowledge from a single firm by transferring and translating knowledge across organizational boundaries [13]. For example, online collaboration tools facilitate information sharing more efficiently and CAD tools enable team members to develop shared meanings of ideas for designing digital innovation [33].

Our findings also suggest that IOS tools and boundary spanners play co-specialized roles in the ideation tasks of enabling digital innovation. Specifically, the effects of IOS tools are augmented during the idea implementation stage because digital artifacts are suitable for transferring codified, explicit knowledge by helping collaborating firms overcome syntax and semantic boundaries [9]. The effects of project coordinators are augmented during the idea generation stage because transferring tacit knowledge relies on human-to-human interactions to help overcome pragmatic boundaries [13,27].

To support these insights, we performed post hoc analysis to test 1) the interaction effects of IOS and the interdependence of idea generation tasks and 2) the interaction effects of boundary coordinators and the interdependence of idea implementation tasks. The results show that these interaction effects do not hold, further strengthening our findings that the role of boundary objects and spanners in knowledge integration depends on the tasks that are associated with tacit or explicit knowledge. Specifically, explicit knowledge integration relies on the information medium that transfers and shares information between teams [25]. Therefore, when idea implementation tasks are present, the effects of IOS tools on integrating explicit knowledge are increased. Tacit knowledge integration relies on team members' willingness and trust to collaborate and project coordinators can help build reciprocal relationships within the team [26,72]. Therefore, when idea generation tasks are present, the effects of boundary spanners on integrating tacit knowledge are increased.

6.2. Theoretical contributions

Our research contributes to the following literature streams. First, this study contributes to the IT-based value co-creation literature by providing a boundary-spanning mechanism to explain how to use IOS tools and project coordinators to engage in different stages of ideation tasks to integrate knowledge across organizational boundaries. Prior research has not developed a systematic understanding of how these factors interact to impact value co-creation. Our boundary-spanning mechanism specifically contributes to the knowledge perspective of IT-based value co-creation by explaining the role of IOS tools and project coordinators in integrating knowledge at an inter-firm digital innovation project [21].

Furthermore, a key finding of this study is the moderating effects of task interdependence. Prior research suggests that high-level interdependencies positively moderate the effects of IT on value co-creation, but has not addressed how the effects may vary across different types of tasks [15]. Our research advances this discourse by examining two types of ideation tasks in digital innovation development. Our findings demonstrate that the interdependence of idea implementation tasks only moderates the effects of IOS tools on knowledge integration, whereas the interdependence of idea generation tasks moderates the effects of project coordinators on knowledge integration. We, therefore, suggest that IOS tools and project coordinators play a co-specialized role in integrating knowledge. Specifically, IOS tools are important for explicit knowledge integration by sharing information to overcome syntax and semantic knowledge boundaries, whereas project coordinators are important for implicit knowledge integration by developing reciprocity among innovation actors to overcome pragmatic knowledge boundaries [13].

Second, our research contributes to the IOS literature by addressing how IOS tools and organizational resources enable knowledge integration for digital innovation. The literature on IOS-based business value predominantly focuses on the role of IOS in information sharing at the operational level [8,7]. Our research suggests that IOS tools alone

cannot lead to digital innovation at the interorganizational level. Their performance impacts are realized through knowledge integration. Moreover, firms need to use the IOS tools in conjunction with co-specialized resources such as project coordinators to realize inter-firm knowledge integration. Specifically, IOS tools are more important when the knowledge barriers in converting ideas to technical solutions are high, and project coordinators are more important when the knowledge barriers in generating ideas are high.

Third, prior research mainly examines the enablement of digital product innovation at a single-firm level. Among these studies, value is captured using market-facing or financial metrics of digital innovation [6]. Our research advances the development of the digital innovation literature by examining how digital innovation is enabled across firms. Research suggests that the integration of domain-specific knowledge is crucial for digital innovation at the inter-firm level [11,46]. However, the understanding of the antecedents of cross-boundary knowledge integration is currently lacking. Our research addresses this issue by demonstrating how boundary objects, spanners, and ideation tasks interact to integrate cross-domain knowledge for digital innovation. Furthermore, different from prior research focusing on financial performance for value realization, our research examines the quality of digital innovation in new features and technical capabilities.

6.3. Practical implications

Our research offers a number of insights for practitioners to participate in collaborative innovation. First, collaborative firms need to invest in both technical and human resources to enable digital innovation at the inter-firm level. IOS tools and project coordinators play different roles in facilitating knowledge integration activities across firms. IOS tools facilitate knowledge integration by developing information in a common format with shared structure and meanings. Project coordinators facilitate knowledge integration by resolving the conflicts that arise from collaboration and by building reciprocity and trust between collaborating firms.

Second, practitioners need to invest in boundary-spanning resources for ideation tasks. Our findings suggest that IOS and project coordinators are co-specialized assets during the ideation process. As human-to-human interactions are vital for developing tacit knowledge, collaborative organizations need to encourage project coordinators to actively engage in the collaboration process especially when the interdependence of tasks is high. When implementing ideas, information for enabling digital innovation becomes codified and explicit. Collaborative organizations, therefore, need to effectively use IOS to transfer explicit knowledge.

Our research further suggests that collaborative organizations may avoid a high level of investment in IOS tools at the idea generation stage of enabling digital innovation. Because our research findings demonstrate that IOS tools become more important during the idea implementation stage, collaborating firms can keep a necessary level of investment in IOS tools at an early stage of the innovation and increase the investment when idea implementation tasks are dominant in the collaboration. By doing so, firms can spare resources to invest in project coordinators at the early stage as they are important for articulating the shared version of the project and for building reciprocity and trust among innovation workers from different organizations.

6.4. Limitations and future research

Our research has several limitations, which provide opportunities for future research. First, we identify R&D and product managers who are in charge of a collaboration project to be the most knowledgeable informant following the key informant approach. Prior research has suggested that some project coordinators can emerge from work practices and may not be the individuals who are nominated at the start of the project [26]. In our research, designated boundary-spanning project

managers oversee the entire project, so they are the most informative respondents to answer our survey questions. Furthermore, single informants may generate response biases, although our assessment shows that common method variance is not a major concern of our research findings. Future research could use a dyadic design to collect data from pairs of project coordinators, one in each collaborating firm [31].

Second, we expect our boundary-spanning mechanisms to be extended to other business contexts. We consider the interdependence of ideation tasks to be critical in boundary-spanning processes for digital innovation. In other business contexts such as supply chain management, future research may examine the interdependence of decision-making or other business practices as boundary-spanning tasks. Furthermore, the four industries we selected may not fully represent the firms in general that participate in interorganizational digital innovation projects. We encourage future research to test our model in other industries and extend the model by adding industry-specific factors.

Third, our study used a cross-sectional research design to measure constructs at a point in time through the survey research method. Although the survey method provides a high level of generalizability, it offers limited in-depth, context-based explanations. For example, idea generation and implementation can be iterative during the process of enabling digital innovation. Future research could use the case study research method to explore how IOS tools and project coordinators are used in the iterative processes of ideation.

Apart from methodological opportunities, future research can also use alternative theoretical perspectives to investigate how to co-create digital products at interorganizational level. For example, prior research suggests that governance is important for the management of inter-firm collaboration [21,73]. Future research could use IT governance theory to examine the influence of different governance mechanisms (e.g., relational governance; structural governance; procedural governance) on innovation outcomes [74].

CRedit authorship contribution statement

Yunfei Shi: Methodology, Formal analysis, Writing – original draft, Writing – review & editing. **Tingru Cui:** Methodology, Formal analysis, Writing – original draft, Writing – review & editing. **Sherah Kurnia:** Methodology, Writing – review & editing.

Appendix A Interview Details

We conducted interviews with key stakeholders to validate and adapt the measurements of IOS tools we identified from prior literature. Based on our targeted industries, We contacted five firms to explore their use of IOS tools in collaborative innovation activities. Three firms agreed to participate in the project. We then conducted interviews with three project managers at each firm. In total, we conducted nine interviews. Each interview lasted for around 45–60 minutes. We recorded and transcribed each interview to ensure the reliability of the information. We asked the interviewees to identify a collaborative innovation project that had been recently completed and to describe the IOS tools they used for the project, and their roles in the project they identified.

Appendix B Assessment of Common Method Variance

Our research is subject to the threat of common method variance because we collected the survey data using a single-informant approach [75]. To mitigate the threats, we kept respondents uninformed about the constructs that we were measuring. We also offered respondents the choice of answering questions anonymously to reduce respondents' evaluation apprehension and to minimize social desirability bias.

We used two approaches to assess potential common method variance in our research. First, we conducted Harmon's one-factor test by entering all the constructs into a principal components factor analysis [75]. One constrained factor accounted for 35.6% of the variance (<

Table B1

Comparison of the Baseline Model and the Model with the Marker Variable.

Path	Baseline Model			Model with Marker		
	Est.	S.E.	Sig.	Est.	S.E.	Sig.
IKI→DIQ	0.595	0.058	0.000	0.595	0.060	0.000
BS→IKI	0.163	0.067	0.014	0.149	2.138	0.033
BO→IKI	0.280	0.059	0.000	0.288	4.942	0.000
BS*IGTI→IKI	0.229	0.058	0.000	0.227	3.877	0.000
BO*ITTI→IKI	0.284	0.122	0.020	0.293	2.145	0.032
Marker→IKI				0.078	1.236	0.217
Marker→DIQ				0.031	0.458	0.647

50%). Second, we applied Rönkkö and Ylitalo's [76] six-step marker variable approach. There were no differences between the baseline model and the model with the marker variable, while all paths saw no change in the level of statistical significance (Table B1). Therefore, common method variance is not a major concern in our research.

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